

**Limited Discovery Power Primarily Drives Poor Transferability of European-Derived
Type 2 Diabetes Polygenic Risk Scores to Central and South Asian Populations**

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BMI 6313: Scientific Writing in Healthcare

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August 5, 2026

Source of data: Pan-UK Biobank genome-wide association study summary statistics and the 1000 Genomes Project Phase 3 reference panel (publicly available, de-identified).

Submission for Publication: This paper is being submitted as a course assignment and is not intended for publication at this time.

Abstract

Polygenic Risk Scores for type 2 diabetes are trained predominantly on European-ancestry genome-wide association studies, despite Central and South Asian populations bearing the highest disease burdens. It remains unclear whether poor cross-ancestry performance reflects divergent genetic architecture, technical artifacts, or insufficient discovery power in non-European cohorts. We conducted a secondary analysis of European and South Asian genotype and summary statistic data, applying clumping and thresholding and Ldpred2-inf under ancestry-matched and mismatched reference panels. Score distributions, decile enrichment, cross-score concordance, locus-level effect heterogeneity, and genome-wide genetic correlation were compared across four score configurations in 478 South Asian and 503 European individuals. European-derived scores shifted South Asian distribution upward by 1.27-1.32 standard deviations, producing 3.8-3.90-fold enrichment in the top risk decile, while concordance between European- and Central and South Asian-derived scores was negligible ($r=-0.001$ to 0.047). Genome-wide genetic correlation was indistinguishable from unity ($r_g=1.27$, $p=6.7\times 10^{-11}$), and none of 8,789 tested loci showed significant effect heterogeneity after multiple-testing correction. Directional concordance of effect estimates rose from 63.7% at low statistical power to 100% at high power. These findings show that miscalibration in Central and South Asian populations stems mainly from limited discovery sample power rather than fundamental differences in disease architecture. Computational improvements alone are unlikely to close this gap. Expanding genome-wide association studies in underrepresented populations is needed to achieve equitable type 2 diabetes risk prediction.

Keywords: *polygenic risk score, cross-ancestry transferability, type 2 diabetes, health equity, genome-wide association study*

Introduction

Polygenic risk scores (PRS) estimate an individual's inherited disease risk from variants identified in genome-wide association studies (GWAS), offering a route to precision medicine approaches such as identifying high-risk individuals before disease onset. For Type 2 Diabetes (T2D), projected to affect 783 million people globally by 2045, this potential is substantial (Hossain et al., 2024). However, clinical utility is severely limited by a persistent representational bias in research: 91% of GWAS participants are of European (EUR) ancestry (Fitipaldi & Franks, 2022). This creates a troubling disparity because the populations most vulnerable to T2D are the least studied and validated for these tools. Earlier identification of high-risk individuals could support timely lifestyle or pharmacologic prevention, reducing downstream complications such as cardiovascular and kidney disease. This study is a secondary analysis of publicly available genomic data rather than new data collection.

Two mechanisms explain why EUR-derived PRS commonly fail elsewhere. First, the frequency of disease-associated variants differs across ancestry groups, so PRS calibrated on EUR data can misrepresent genetic liability in other populations (Mahajan et al., 2022). Second, genetic variants are inherited in clusters called haplotypes, and the patterns of these clusters, known as linkage disequilibrium (LD) structure, differ significantly across populations. Because PRS use index variants as proxies for nearby variants, LD mismatch can cause the index variants to miss actual causal variants elsewhere. These two factors account for 70-80% of the cross-ancestry accuracy loss, making them primary targets for understanding transferability failure (Kachuri et al., 2024; Wang et al., 2020).

Despite this literature, critical gaps remain for Central and South Asian (CSA) populations, who carry the highest global T2D burden yet are among the most underrepresented

in GWAS (Ding et al., 2023, Sattar et al., 2020). No prior study has comprehensively evaluated whether cross-ancestry T2D PRS failure in CSA populations reflects fundamental architectural differences, correctable technical artifacts, or insufficient discovery sample power in CSA cohorts. Understanding which mechanism dominates has direct implications for whether future advances require computational innovation, methodological refinement, or expanded population-specific GWAS discovery. We hypothesized that limited CSA GWAS discovery power, rather than divergent genetic architecture or technical/computational artifacts, would be the dominant driver of poor cross-ancestry PRS transferability in this population.

Two widely used PRS methods are clumping and thresholding (C+T) and LDpred2-inf. C+T selects a set of near-independent variants based on LD and association significance, then sums their weighted effect sizes. Because C+T typically relies on EUR-estimated LD, its performance can decline when applied to populations with different LD patterns (Choi et al., 2020). LDpred2-inf instead estimates the effects of all variants simultaneously while modeling genome-wide LD, making it better suited to highly polygenic traits such as T2D. However, its accuracy still depends on how well the LD reference panel matches the target population (Privé et al., 2021). Comparing these methods allows assessment of whether more sophisticated LD modeling can improve cross-ancestry transferability.

To address these gaps, we evaluated the cross-ancestry transferability of EUR-derived T2D PRS to CSA individuals using two established polygenic scoring methods and ancestry-matched versus mismatched genetic reference panels to isolate distinct sources of transferability failure. Our analytical approach employed a hierarchical framework examining the problem at multiple biological levels: population-level (score distributions and clinical risk stratification), locus-level (individual variant effect heterogeneity), and genome-wide (overall genetic

correlation and shared biology) (Bulik-Sullivan et al., 2015; Ding et al., 2023; Kachuri et al., 2024; Mahajan et al., 2018).

Our results show that these prediction errors in CSA individuals stem mainly from allele-frequency differences and a shortage of CSA-specific GWAS data, not from fundamental differences in diabetes biology between populations. This conclusion is supported by the high overall genetic correlation between EUR and CSA groups. These findings indicate that the primary barrier to equitable T2D risk prediction is a lack of CSA data, and that closing this gap will require expanded CSA GWAS efforts rather than improved computational methods alone. Without CSA-specific genomic data, PRS-based screening could under-estimate or over-estimate diabetes risk in this population, delaying prevention efforts where they are needed most (Martin et al., 2019).

Methods

Data Sources and Phenotype Definition

This study employed a cross-ancestry PRS evaluation framework to assess the transferability of a EUR-derived T2D PRS to SAS individuals using publicly available data. EUR and CSA GWAS summary statistics for T2D (ICD-10: E11) were obtained from the Pan-UK Biobank (Pan-UKB) release (Karczewski et al., 2025). Individual-level genotype data were obtained from the 1000 Genomes Project (1KGP) Phase 3 reference panel, restricted to the EUR and SAS populations, which served as ancestry-matched genotype proxies for the EUR and CSA discovery samples (The 1000 Genomes Project Consortium et al., 2015). 1KGP was selected because it is among the few public resources offering individual-level, ancestry-labeled genotypes for European and South Asian populations at adequate sample sizes.

Quality Control

Summary statistics were filtered using Pan-UKB's low-confidence flag (allele count ≤ 3 or combined minor allele count ≤ 20) and by removing strand-ambiguous A/T and C/G SNPs, yielding 11,239,095 variants. Target genotype was quality-controlled separately within each population in *PLINK2* (Chang et al., 2015) to avoid confounding allele-frequency filters with population structure, retaining variants with minor allele frequency (MAF) $> 1\%$, genotype missingness $< 5\%$, and Hardy-Weinberg p-value $> 1 \times 10^{-6}$. The intersection of SNPs passing quality control in both populations (1,422,958 variants) was utilized for all subsequent steps. Principal component analysis (PCA) was performed on LD-pruned genotypes ($r^2 < 0.2$, 1000-SNP window) using *bigsnp* (Privé et al., 2021) to verify ancestry assignments and identify outliers (> 6 SD from population centroids) on the top 10 principal components (PCs).

PRS Construction and Standardization: Experimental Design and Tested Conditions

Two independent algorithms were used to construct T2D PRS, allowing assessment of whether transferability findings were specific to a single scoring method. C+T was performed using *PLINK2* (Chang et al., 2015) with LD-based clumping ($r^2 < 0.1$, 250-kb window) applied separately using LD reference panels (Privé et al., 2021). PRS were computed across multiple GWAS p-value thresholds (5×10^{-8} -1.0) using a common clumped SNP set, with $p < 0.05$ used for the primary analysis (Choi et al., 2020). LDpred2-inf was run using ancestry-matched LD reference panels built separately for the EUR and SAS genotype sets (Privé et al., 2021). The model derives posterior effect sizes in closed form using a genome-wide heritability estimate ($h^2 = 0.352$) without requiring the computationally intensive Markov Chain Monte Carlo (MCMC) sampling.

Four score configurations were generated: EUR effect sizes with EUR LD in EUR samples (reference), EUR effect sizes with EUR LD in SAS samples (clinical simulation), EUR

effect sizes with SAS LD in SAS samples (LD-reference sensitivity), and CSA effect sizes with SAS LD in SAS samples (full ancestry-matched comparison). Scores were standardized two ways: within-population (mean=0, SD=1) and against EUR-derived parameters.

Outcome Measures and Statistical Analysis for PRS Evaluation

Because 1KGP lacks phenotypic data, transferability was assessed using genotype-only metrics rather than predictive accuracy. Mean shift quantified the difference, in EUR-standardized SD units, between SAS and EUR score distributions relative to the reference condition. Decile enrichment compared the proportion of SAS individuals in each EUR-defined risk decile against the 10% expected under calibration, with top-decile enrichment as the primary clinical-relevance statistic (Ding et al., 2023).

Cross-score concordance was assessed by Pearson correlation between EUR- and CSA-derived scores within SAS individuals. At EUR genome-wide significant loci ($p < 5 \times 10^{-8}$), directional concordance was evaluated by comparing the signs of EUR and CSA effect estimates and stratifying results by CSA absolute Z-score magnitude to distinguish true heterogeneity from power-related noise. Locus-level heterogeneity was assessed using Cochran's Q statistic ($Q = (\beta_{EUR} - \beta_{CSA})^2 / (SE_{EUR}^2 + SE_{CSA}^2)$), which tests whether EUR and CSA effect sizes differ beyond that expected by chance. Significance was determined using the chi-squared distribution with Bonferroni correction ($\alpha = 0.05/n_{loci}$). Genome-wide genetic correlation (r_g), which measures the extent to which genetic effects are shared between populations, and SNP-heritability (h^2), which determines the expected overall genetic contribution to the trait, were estimated by LDSC (Bulik-Sullivan et al., 2015; Levin, 2022) to contextualize any observed score miscalibration against shared versus divergent biology.

Results

Quality control and population structure recapitulate expected ancestry stratification. Genotype quality control retained 503 EUR and 484 SAS samples typed at 1,422,958 variants shared between populations. PCA cleanly separated the two populations along PC1, which explained 50.3% of total variance (Figure 1), and six SAS individuals exceeding six standard deviations from the SAS centroid were excluded as outliers, leaving 478 SAS samples for all downstream scoring. No EUR samples were flagged.

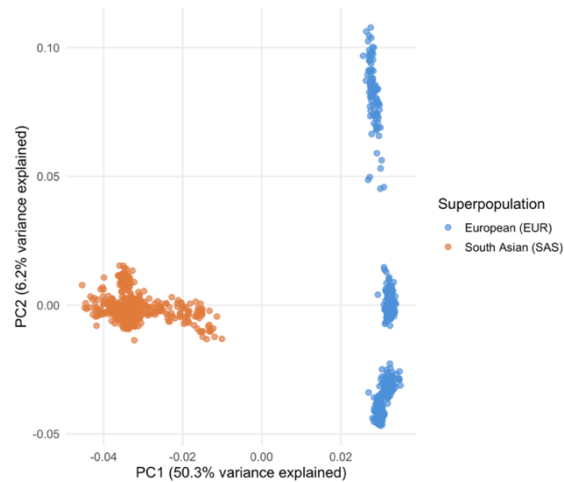


Figure 1. PCA Between EUR and SAS. Scatter plot of PC1 and PC2 derived from genotype data, with each point representing one individual colored by self-reported superpopulation.

Under both scoring methods, EUR-derived PRS distributions shift substantially when applied to SAS individuals. When EUR effect sizes drove the score, the SAS distribution is shifted rightward relative to EUR as the C+T score averaged 1.32 SD higher in SAS and the Ldpred2-inf score showed a nearly identical shift (1.27 SD) (Figures 2A & 2D). Substituting a SAS-specific LD reference while retaining EUR effect sizes narrowed the shift to 1.19 SD (Figure 2B). When CSA-derived effects were used, the SAS distribution is displaced far to the right (5.55 SD), and its shape is substantially wider (3.83 SD vs. 1.03 for EUR) (Figure 2C).

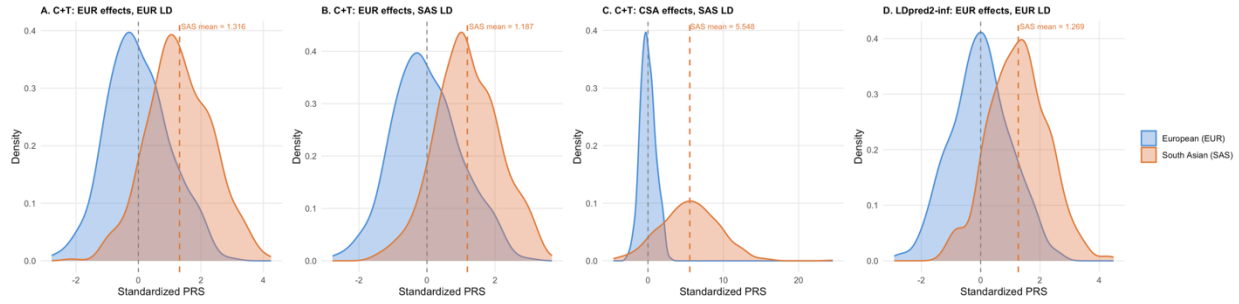


Figure 2. Density Distributions of Standardized PRS in EUR and SAS Individuals. (A) C+T, EUR effect sizes with EUR LD reference (reference configuration). (B) C+T, EUR effect sizes with SAS LD reference. (C) C+T, CSA effect sizes with SAS LD reference. (D) LDpred2-inf, EUR effect sizes with EUR LD reference. Gray dashed line marks the EUR mean (0); orange dashed line marks the SAS mean, with the value labeled on each panel.

Although the CSA-derived LDpred2-inf score remained shifted by 1.29 SD relative to the EUR reference distribution, it was substantially less variable than the EUR-derived score (SD=0.34 vs. 0.98; Figure 3). This compression likely reflects the LDpred2-inf's heritability-based shrinkage, whereas the lower heritability estimated from the CSA GWAS produced stronger shrinkage of effect sizes toward zero.

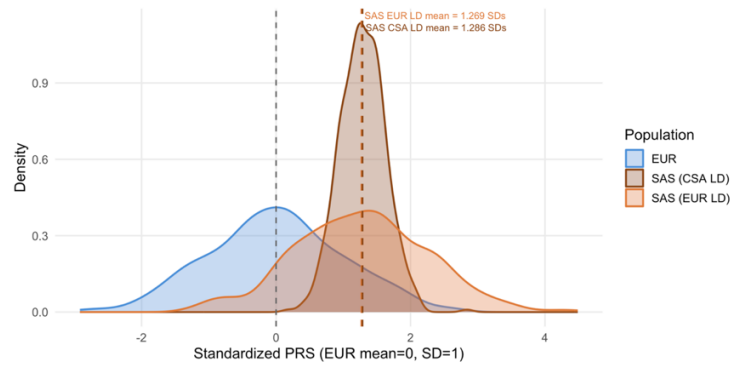


Figure 3. Density Distribution of LDpred2-inf PRS Across Ancestry Groups. Overlapping density curves of LDpred2-inf standardized PRS using either EUR LD reference (orange) or CSA LD reference (dark brown). Dashed vertical lines mark each SAS distribution's mean.

Decile analysis showed pronounced upward redistribution of SAS individuals across both scoring methods. Relative to 10% expected under calibration, SAS individuals were nearly absent from the lowest EUR-defined risk deciles and approximately fourfold enriched in the highest decile (C+T: 3.90-fold, LDpred2-inf: 3.82-fold) (Figure 4).

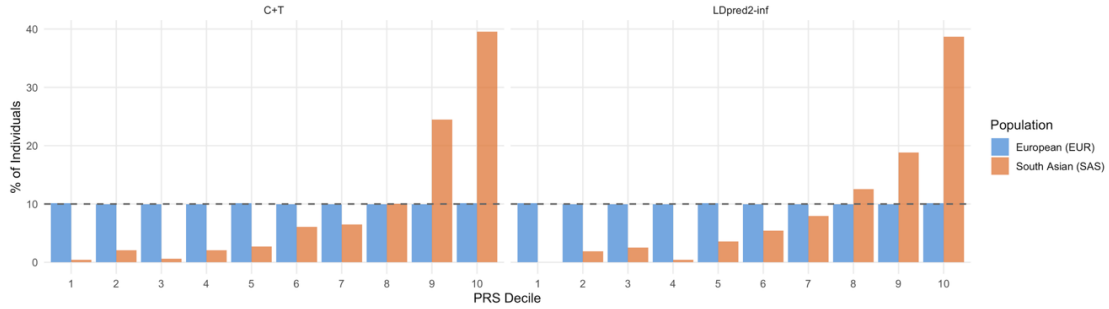


Figure 4. PRS Decile Distribution. Bar charts showing the percentage of individuals falling into each of ten PRS deciles (1=lowest risk, 10=highest risk), separately for C+T (left panel) and LDpred2-inf (right panel). Decile boundaries were defined using the EUR distribution. The dashed horizontal line at 10% indicates the proportion expected under perfect calibration (equal representation across deciles).

We next asked whether EUR-derived and CSA-derived scores agree at the individual level within 478 SAS participants. Scatterplots showed no relationship for C+T ($r=-0.001$; Figure 5A) and only a very weak positive relationship for LDpred2-inf ($r=0.047$; Figure 5B). This near-zero correlation means that a person ranked high-risk by the European-derived score and one ranked high-risk by the CSA-derived score are, in practice, not the same people.

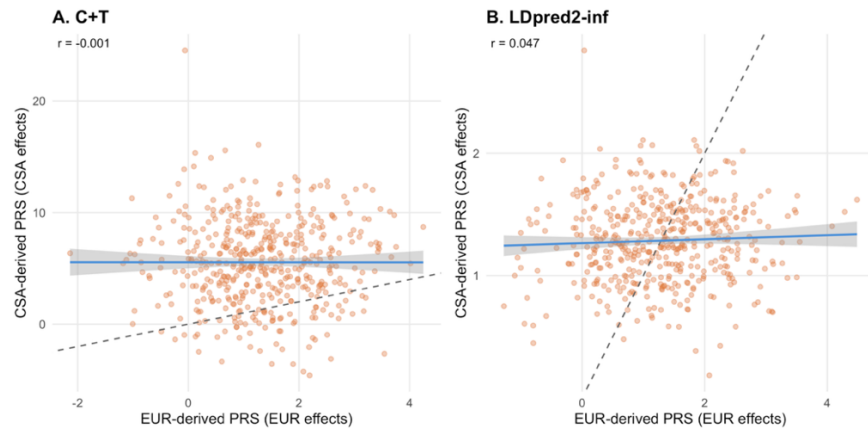


Figure 5. Cross-Ancestry Score Concordance. Each point represents one SAS individual ($n=478$). The blue line is the fitted ordinary least-squares regression line with 95% confidence band (gray shading). The dashed diagonal line indicates perfect concordance (slope=1, intercept=0).

To determine whether the miscalibration reflects biological differences or statistical noise, we examined cross-ancestry genetic architecture. LD score regression estimated similar SNP-heritability for T2D in EUR ($h^2=0.036$) and CSA ($h^2=0.028$, $p=0.49$), suggesting comparable genetic contributions to disease risk. The estimated genetic correlation was high

($r_g=1.27$, $p=6.7 \times 10^{-11}$), indicating that the genetic effects underlying T2D are largely shared between the two populations despite the poor transferability of EUR-derived PRS.

Table 1. Directional Concordance of Effect Estimates Between EUR and CSA Population

CSA $ Z $ bin	N SNPs	% same direction as EUR	95% CI
<1	5,806	63.7	62.4–64.9
1-2	2,342	71.3	69.5–73.1
2-3	439	98.0	96.6–99.3
3-4	81	100	100–100
>4	121	100	100–100

Note. Loci with Bonferroni-significant heterogeneity (Q test): 0 of 8,789 tested.

Table 1 further shows that directional concordance between EUR and CSA effect estimates depends entirely on how well-powered the CSA signal is. Although 63.7% SNPs with $|Z| < 1$ in CSA pointed in the same direction as EUR, concordance reached 97.9-100% among SNPs with $|Z| > 3$ (Table 1). Figure 6 illustrates this pattern as a scatter plot of EUR effect size against CSA effect size for all 8,789 EUR genome-wide significant loci, colored by the degree of EUR-CSA heterogeneity. At loci where the CSA signal is strong enough to estimate reliably, points cluster along the diagonal. At underpowered loci, they scatter around zero on the y-axis, pulling the overall correlation down to $r=0.417$ across all loci versus $r=0.609$ when restricted to loci with $|Z| \geq 1$. CSA effect sizes also appear systematically compressed relative to EUR, as most points fall below the diagonal. Formal testing with Cochran's Q across all 8,789 loci, including TCF7L2 (strongest EUR T2D signal), identified zero variants surviving Bonferroni correction for EUR-CSA heterogeneity.

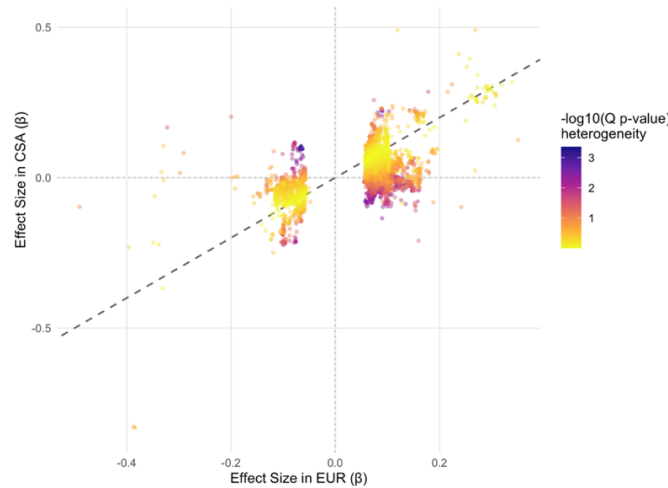


Figure 6. Locus-Level Comparison of Effect Size Estimates. Scatter plot in which each point represents one of the 8,789 EUR genome-wide significant T2D loci. Points are colored by degree of EUR-CSA effect-size heterogeneity, quantified as $-\log_{10}$ of the Cochran's Q p-values (yellow=low heterogeneity; dark purple=high heterogeneity). The dashed diagonal line indicates perfect concordance (slope=1).

Because allele frequency differences between populations can contribute to score miscalibration independently of effect-size estimation, we compared risk allele frequencies at the 8,789 EUR genome-wide significant loci. In Figure 7A, the majority of the points fall close to the diagonal. Figure 7B, however, shows the distribution of absolute frequency differences, revealing that most loci have differences below 0.1 (mean=0.076) with a smaller cluster of loci visible near differences of 0.25-0.30.

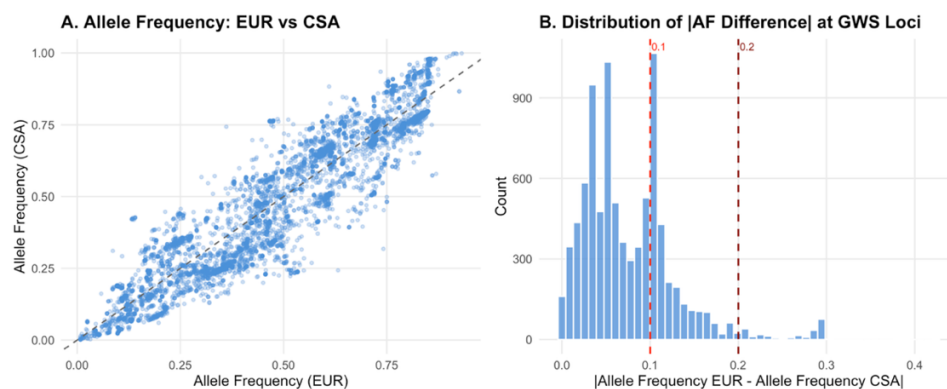


Figure 7. Allele Frequency Difference at EUR GWAS T2D Loci. (A) Scatter plot of risk allele frequency in EUR vs CSA for each locus. The dashed diagonal line indicates equal frequencies across populations. (B) Histogram of the absolute allele frequency difference across all loci. The dashed vertical line mark reference thresholds.

Discussion

This study evaluated whether PRS derived from large EUR GWAS can be validly applied to CSA populations, a question with direct relevance to health equity in genomic medicine. We hypothesized that ancestral differences in LD structure, allele frequencies, and effect-size magnitudes would reduce PRS transferability in CSA populations. Our findings indicate that although EUR-derived PRS are substantially miscalibrated when applied to CSA individuals, this miscalibration primarily reflects limited CSA GWAS discovery power together with population differences in allele frequencies and LD structure, rather than fundamentally different architectures.

Genome-wide analysis supports a largely shared genetic basis for T2D between EUR and CSA populations. The estimated genetic correlation was high ($r_g=1.27$, $p=6.7\times 10^{-11}$), and none of 8,789 genome-wide-significant loci, including the most robustly replicated T2D signal (TCF7L2), showed significant heterogeneity after multiple-testing correction. Nevertheless, together with the absence of significant locus-level heterogeneity ($h^2=0.028$, $p=0.49$), these results are consistent with broadly shared disease biology rather than population-specific genetic mechanisms. Overall, the shared genetic architecture and absence of locus-level heterogeneity is consistent with Mahajan et al. (2022) and Ding et al. (2023), who similarly found that T2D risk loci are largely conserved across ancestries rather than population-specific.

The locus-level analysis further supports this interpretation. Directional concordance increased from 63.7% among weak CSA associations ($|Z| < 1$) to 97.9-100% among well-powered loci ($|Z| > 3$), indicating that agreement improved as effect-size estimates become more precise rather than due to a shift in the underlying biology architecture (Table 1). CSA effect sizes were also systematically attenuated relative to EUR estimates (Figure 6), consistent with reduced statistical power from the substantially smaller CSA GWAS (1,662 cases) (Dudbridge,

2013). The lack of significant Cochran's Q results should therefore not be interpreted as definitive evidence of complete homogeneity but rather as reflecting limited power to detect modest heterogeneity.

Allele frequency divergence and LD mismatch contributed independently to PRS miscalibration but explained only part of the observed transferability loss. Although most EUR risk loci had similar allele frequencies across populations, a subset showed substantial divergence (0.25-0.3; Figure 7B), and replacing the EUR LD reference with an ancestry-matched SAS reference reduced the C+T mean shift from 1.32 to 1.19 SD (Figure 2A-B). The magnitude and pattern of score miscalibration we observed is consistent with Kachuri et al. (2024) and Wang et al. (2020), who attributed most cross-ancestry PRS degradation to discovery-power and LD-reference mismatch rather than divergent biology.

The practical consequences of this miscalibration are substantial. EUR-derived scores shifted the entire CSA score distribution upward by approximately 1.3 standard deviations under both scoring methods (C+T: +1.32 SD; LDpred2-inf: +1.27 SD), producing nearly fourfold enrichment of CSA individuals in the highest EUR-defined risk decile (C+T: 3.90-fold enrichment, LDpred2-inf: 3.82-fold enrichment) (Figure 4). Such a pattern could be misinterpreted as evidence of markedly elevated inherited risk when it instead reflects systematic score miscalibration. Consistent with this interpretation, CSA-derived effect sizes produced unstable C+T scores because of noise amplification (SD=3.83; Figure 2C), whereas LDpred2-inf generated an overly compressed score distribution through stronger heritability-based shrinkage (SD=0.34 vs. 0.98). Near-zero concordance between EUR- and CSA- derived scores ($r=-0.001$ and 0.047) further indicates that neither approach currently provides reliable individual-level risk stratification in this population (Figure 5). Although LDpred2-inf incorporates more

sophisticated LD modeling than C+T, it did not improve transferability, suggesting that the principal limitation is the composition and size of available discovery cohorts rather than the scoring algorithm itself.

Several limitations qualify these conclusions. The 1000 Genomes panel lacks phenotype data, so transferability was assessed through genotype-level metrics rather than predictive accuracy. The poor performance of the CSA-based PRS is largely due to the small CSA GWAS sample size (1,662 cases), making underrepresentation of non-European populations a key finding rather than just a limitation. CSA and SAS also collapse population structure within South Asia not fully captured by the 1KGP SAS population.

Future work should prioritize expanding CSA-specific GWAS and larger phenotype cohorts to directly evaluate clinical PRS performance. Existing efforts (e.g., DIAMANTE, SAT2D, the South Asia Biobank) could substantially increase CSA case numbers beyond the 1,662 available in this study. Because miscalibration reflects limited ancestry representation, not distinct biology, expanding CSA inclusion offers the most effective path to equitable T2D PRS.

Acknowledgements

I thank my classmates of BMI 6313: Scientific Writing in Healthcare at the McWilliams School of Biomedical Informatics, UTHealth Houston, for their feedback and peer review of earlier drafts of this manuscript. Their comments improved the clarity and statistical framing of this work. I also acknowledge the Pan-UK Biobank and 1000 Genomes Project consortia for making the summary statistics and genotype data used in this analysis publicly available.

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